

Correct Answers, Uneven Grounding

The distributed-evidence gap in legal RAG

-12.2 pp

final page-grounding gap

95% CI: -19.0 to -5.9

Answer scores stay high. Complete source support falls when evidence spans pages.

95 eligible questions / 30 PDFs / 590 pages

1 HYPOTHESIS

Correct answers and complete citation sets are different outcomes. Aggregate scores can hide whether their mismatch depends on evidence shape.

RESEARCH QUESTION

How does distributed evidence affect answer quality and page grounding in a legal RAG pipeline?

H1

Multi-page evidence will preserve answer quality but weaken final grounding; uniform narrowing may widen the gap.

59

36

● localized

◆ distributed

exactly one gold page

multiple gold pages

Answer quality / mean benchmark score on a 0-1 scale.

Source grounding / page F-beta, beta = 2.5, weighting recall.

Evidence shape / 35 of 36 distributed questions also span documents; the effects are confounded.

2 METHODOLOGY

01 Page-first ingestion 30 DIFC PDFs / 590 pages	02 Multi-view index Page / section / clause / table	03 Hybrid retrieval BM25 + Qwen3 Embedding	04 Fusion + reranking RRF + Qwen3 Reranker	05 Typed answering gpt-5.4-mini + validation	06 Page attribution Supported pages + trace
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Page identity is preserved from ingestion to emitted citations.

● Localized: one gold page ◆ Distributed: multiple gold pages

One saved pipeline run; architecture explains traceability, not causality.

3 RESULTS

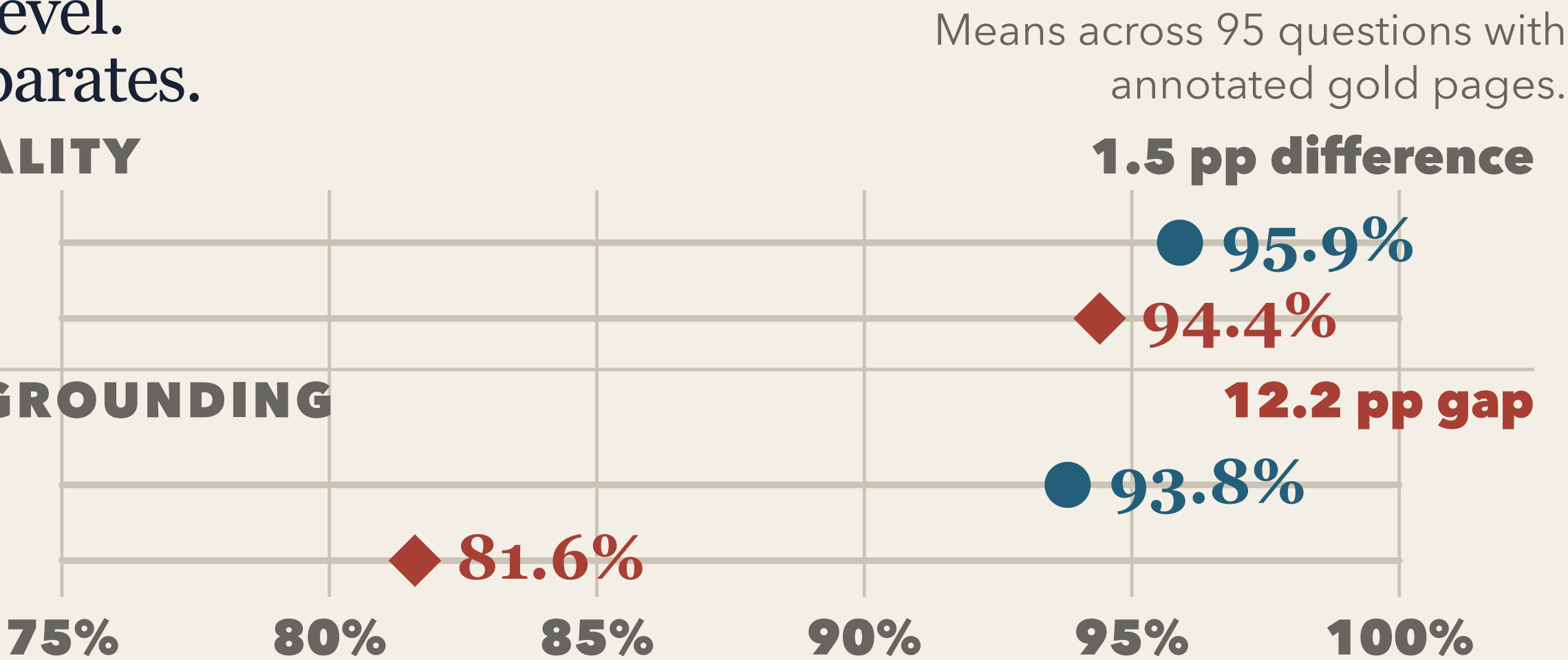
Answers stay level.
Grounding separates.

ANSWER QUALITY

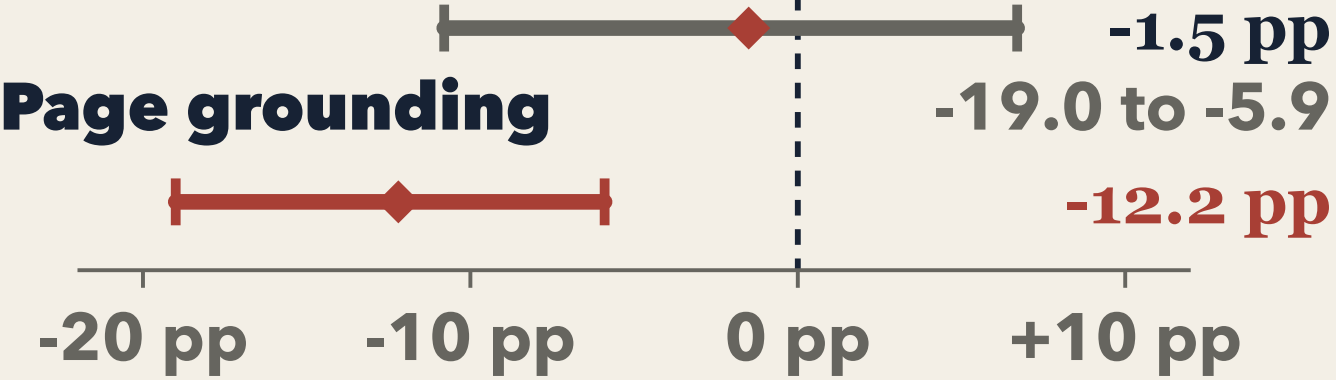
Localized
Distributed

FINAL PAGE GROUNDING

Localized
Distributed



DISTRIBUTED - LOCALIZED / 95% CI
Answer score -10.8 to +6.7



20,000 stratified bootstrap resamples. The answer interval crosses zero; the grounding interval does not.

PERFECT ANSWER, GROUNDING BELOW 0.8

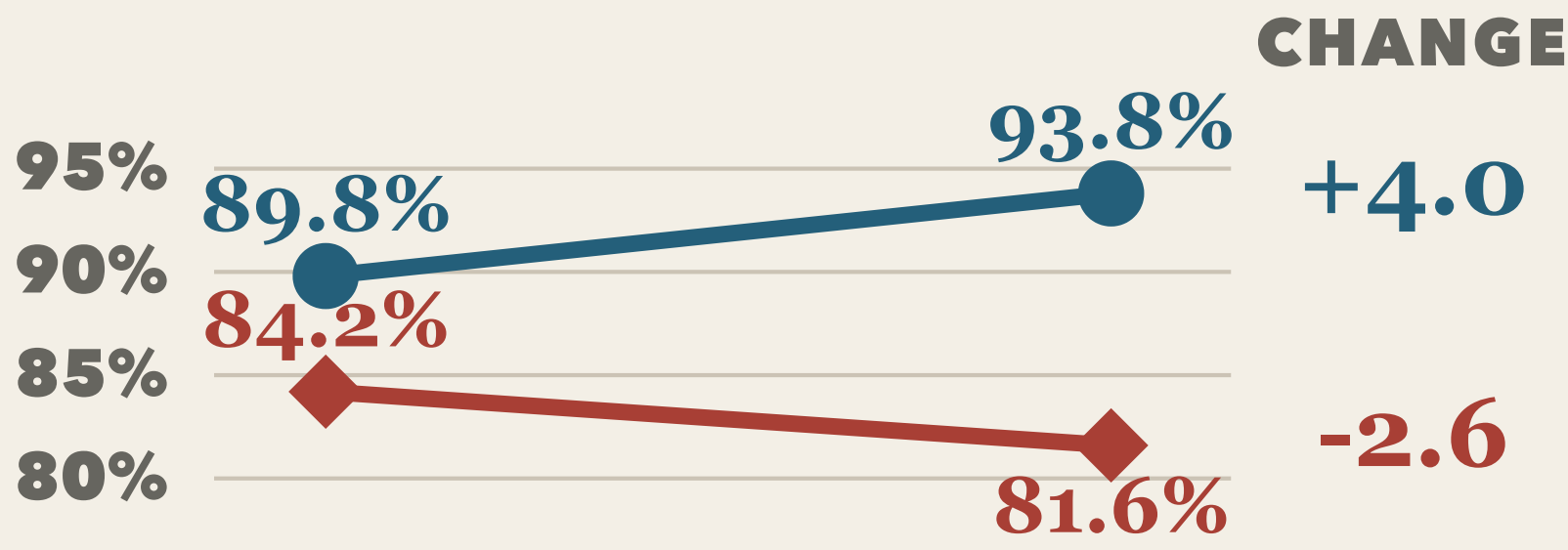


A correct-looking answer is not enough: incomplete support is much more common when gold evidence spans pages.

4 DIAGNOSTIC

Citation narrowing moves groups apart.

Page F-beta before and after answer-conditioned attribution, from the same run.



RAW RETRIEVAL FINAL ATTRIBUTION

DESIGN IMPLICATION

Use evidence spread as a routing signal: narrow localized support; preserve and aggregate distributed support.

Gold evidence shape is unavailable at inference time. Observable proxies include retrieval spread, cross-document dependencies, and claim-level support checks.

LIMITATIONS

One saved development run; retrospective and observational analysis; page and document spread are confounded; answer types and scoring methods are uneven; the grounding-below-0.8 threshold is descriptive.

INTERPRETATION BOUNDARY

The evidence-shape pattern is an association, not a causal effect. Raw-to-final changes occur inside one pipeline run and are not an ablation of independently executed systems.

REPRODUCIBILITY

github.com/SteinVR/Agentic-RAG-Challenge
20,000 stratified bootstrap resamples / seed 20260903.
Configuration, answer scores, and the page-level grounding ledger are retained with the saved run.